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CASE STUDY

Topic : Work Study at an Automobile Assembly Plant

S.No	Description	Maximum Mark	Marks Awarded
1.	Problem Understanding	10	
2.	Innovation & Creativity	20	
3.	Presentation, Communication & Response for Queries	10	
	Total Marks	**40**	

Subject Incharge

1. Title

Work Study at an Automobile Assembly Plant:

A Comprehensive Application of Method Study and Time Study Techniques to Eradicate Bottlenecks and Improve Productivity at Trident Motors

2. Executive Summary

This comprehensive operations management case study examines the production challenges at the Trident Motors Assembly Division, a mid-sized automobile manufacturing plant responsible for assembling popular small passenger hatchbacks. Despite operating on a rigorous three-shift schedule where the workforce appears continuously occupied, the facility has consistently failed to meet its production target of 260 vehicles per shift, plateauing at an average of only 205 vehicles. This 21% production deficit is accompanied by uneven work pacing, severe bottlenecks, high operator fatigue, and increased absenteeism.

To diagnose and resolve these systemic inefficiencies, an exhaustive Work Study was conducted on the plant's trim and final assembly line. Utilizing the core branches of Work Study—Method Study and Time Study—the investigation revealed that the perceived line speed issues were actually the result of an unbalanced workload. Specifically, the Wiring Harness and Dashboard Fitting stations were identified as critical bottlenecks, requiring 116 seconds of standard time against a strict conveyor cycle time of 92 seconds.

The recommendations formulated in this report bypass capital-intensive machinery upgrades in favor of low-cost, high-impact ergonomic and spatial reorganizations. By redesigning workstation layouts, introducing mechanical support jigs, pre-kitting complex parts, and rebalancing minor work elements to under-utilized stations, Trident Motors can bridge the 24-second cycle time gap. This document provides a complete theoretical framework, granular data analysis, standard time calculations, and a practical implementation roadmap to guide the facility back to its targeted output while simultaneously reducing physical strain on its human workforce.

3. Introduction

3.1 The Philosophy of Work Study

Work Study is a foundational management technique utilized to meticulously examine the way human work is performed within an organization. Its primary objective is to systematically investigate all factors affecting the efficiency and economy of a situation, with the ultimate goal of finding the easiest, safest, and most effective method of performing a task.

Furthermore, it seeks to determine the standard time it should take a qualified, trained worker to complete that task under normal conditions.

The discipline evolved from the pioneering scientific management theories of Frederick W. Taylor, who advocated for the precise measurement of work, and was later refined by Frank and Lillian Gilbreth, who introduced the concept of motion study to eliminate wasted human effort. In modern manufacturing, Work Study remains an indispensable tool for maximizing resource utilization without relying solely on exhaustive physical labor.

3.2 The Automobile Assembly Context

Automobile assembly plants represent the classic, quintessential setting for the application of Work Study principles. In these facilities, a single vehicle chassis passes through dozens of sequential workstations, each dedicated to fitting a specific set of components—from heavy engines to delicate wiring harnesses. Because these lines are typically conveyor-paced (moving at a continuous, fixed speed), the synchronization of tasks is paramount.

If one workstation takes noticeably longer than its designated time allowance, it acts as a constraint or "pace-setter" for the entire facility. Downstream workers are left waiting for the delayed vehicle, while upstream workers must halt to prevent a collision of inventory. Consequently, even a minor imbalance of a few seconds at a single station can compound over an eight-hour shift, completely derailing the plant's daily output goals.

3.3 Introduction to Trident Motors Assembly Division

Trident Motors Assembly Division was established eight years ago as a specialized assembly hub for a highly popular hatchback model. Operating downstream in the supply chain, the plant does not fabricate major components; it receives pressed body shells, pre-assembled engines, and various sub-components from external vendors. The facility's sole focus is on sequential assembly, trim fitting, and final quality testing.

The main assembly line is structurally divided into three primary zones:

1. **The Trim Line:** Dedicated to interior fittings, including seats, dashboard consoles, and complex electrical wiring.
2. **The Chassis Line:** Dedicated to heavy mechanical integration, where the engine, transmission, suspension, and exhaust systems are mated to the body shell.
3. **The Final Line:** Dedicated to finishing touches, such as wheel mounting, fluid filling, and final road-testing inspections.

The plant currently employs 320 direct assembly line workers, distributed across 42 distinct workstations. Management set a stringent daily target of 260 vehicles per shift to meet market demand. However, for the past twelve months, the line has severely underperformed, averaging a maximum output of only 205 vehicles per shift.

4. Theoretical Framework and Literature Review

To accurately diagnose the operational failures at Trident Motors, the investigation was grounded in established Operations Management and Industrial Engineering frameworks.

4.1 Method Study

Method Study is the systematic recording and critical examination of existing and proposed ways of doing work. It is designed to develop and apply easier and more effective methods, thereby reducing costs. The standard procedure for Method Study follows the "SREDIM" sequence:

- **Select:** Choose the work to be studied (e.g., bottleneck stations).
- **Record:** Document all relevant facts of the present method using process charts and video analysis.
- **Examine:** Critically challenge the recorded facts (asking Who, What, Where, When, Why, and How).
- **Develop:** Formulate the most practical, economic, and effective method.
- **Install:** Implement the new method as standard practice.
- **Maintain:** Ensure the new standard is routinely followed.

4.2 Time Study and Standard Time

Time Study is a work measurement technique used to establish the time required for a qualified worker to carry out a specified job at a defined level of performance.

- **Observed Time:** The raw time captured via a stopwatch.
- **Normal Time:** The observed time adjusted by a "Performance Rating" (assessing the worker's speed and effort relative to a standard ideal).
- **Standard Time:** The normal time plus necessary "Allowances" (typically 10-15% for personal needs, basic fatigue, and unavoidable delays).

4.3 Line Balancing and Cycle Time

Assembly Line Balancing is the process of assigning tasks to workstations in such a way that the workstations have approximately equal time requirements.

- **Cycle Time:** The maximum time allowed at each workstation to complete its set of tasks on a unit. At Trident Motors, the fixed conveyor speed dictates a Cycle Time of exactly 92 seconds per car.
- **Idle Time:** The productive time wasted when a worker finishes their task faster than the Cycle Time.

5. Problem Statement

The Trident Motors Assembly Division is experiencing a chronic inability to achieve its production quota of 260 vehicles per shift, currently outputting only 205 vehicles. This 21% gap exists despite visual indicators suggesting that the workforce is continuously engaged and rarely standing idle.

Supervisory reports indicate a chaotic work environment: certain workstations regularly fall behind the conveyor pace, forcing operators to rush, skip steps, or pull parts off the main line to finish them manually at the side. This ad-hoc "catch-up" behavior is generating confusion, increasing the risk of safety incidents, and resulting in costly rework at the final inspection stage. Furthermore, operators assigned to the Trim Line—specifically the Wiring Harness and Dashboard Fitting stations—have formally complained of severe physical fatigue and awkward working postures. Human Resources data corroborates these complaints, showing a disproportionately high rate of absenteeism at these specific stations compared to the rest of the facility.

The overarching problem is a lack of data-driven operational visibility. Management is paralyzed by conflicting theories: some supervisors blame a conveyor speed that is "too fast," others blame "slow workers," and some point to supply chain material shortages. Without a systematic, empirical study of the exact work elements being performed, Trident Motors cannot identify the root cause or formulate a targeted solution.

6. Objectives and Scope of the Study

6.1 Primary Objectives

The central aim of this case study is to empirically apply Work Study techniques (Method Study and Time Study) to the trim and final assembly line of Trident Motors. The goal is to identify the true root causes of the line's underperformance and to recommend highly practical, low-capital improvements to achieve the 260-vehicle target.

6.2 Secondary Objectives

- To systematically **record and analyze** the existing method of work at each of the 42 workstations on the assembly line.
- To **measure the actual time** taken to perform each task utilizing stopwatch time studies, and to mathematically compare these standard times against the 92-second conveyor cycle time.
- To clearly **identify bottleneck stations** (over-utilized) and **idle stations** (under-utilized) to facilitate accurate line rebalancing.
- To **recommend ergonomic and spatial changes** to workstation layouts that will eliminate non-value-adding motions and reduce operator fatigue.
- To provide a **replicable mathematical framework** for standard time and line balancing calculations that plant management can utilize for future model launches.

6.3 Scope and Limitations

The scope of this investigation is strictly limited to the internal assembly processes on the Trim, Chassis, and Final lines at Trident Motors. It does not encompass supplier manufacturing processes, external logistics, or the design engineering of the vehicle itself.

7. Research Methodology and Data Collection

To ensure that all conclusions were based on undeniable facts rather than supervisory assumptions, an industrial engineering team spent three weeks on the shop floor utilizing standard Work Study data collection tools.

7.1 Direct Observation and Process Charting

The engineering team physically walked the entire length of the assembly line. For all 42 workstations, they created an outline process chart. This documentation recorded every single element of work, every physical movement of the operator (reaching, bending, walking), and every microscopic delay in the standard sequence.

7.2 Stopwatch Time Study

To gather quantitative data, the team utilized digital stopwatches and standard Time Study sheets. They broke down the tasks at each station into basic elements (e.g., "Pick up wire," "Route through firewall," "Fasten clip"). Repeated time readings were taken across different shifts and different operators to account for normal human variation and to calculate an accurate average observed time.

7.3 Motion Study and Video Analysis

For stations exhibiting high rates of worker complaints (specifically the Wiring Harness and Dashboard Fitting stations), continuous video recording was employed. This allowed the engineering team to review the work frame-by-frame in slow motion, applying Gilbreth's "Therblig" analysis to identify awkward body mechanics, micro-hesitations, and unnecessary hand movements that the naked eye might miss.

7.4 Qualitative Interviews and Records Review

- **Interviews:** Brief, informal interviews were conducted with operators to capture their firsthand experiences regarding difficult tasks, localized parts shortages, and poorly designed tools.
- **Historical Records:** A twelve-month review of production logs, quality rejection reports, and absenteeism records was conducted to verify that the identified bottlenecks were chronic issues, not temporary anomalies.

8. Comprehensive Data Collection and Observations

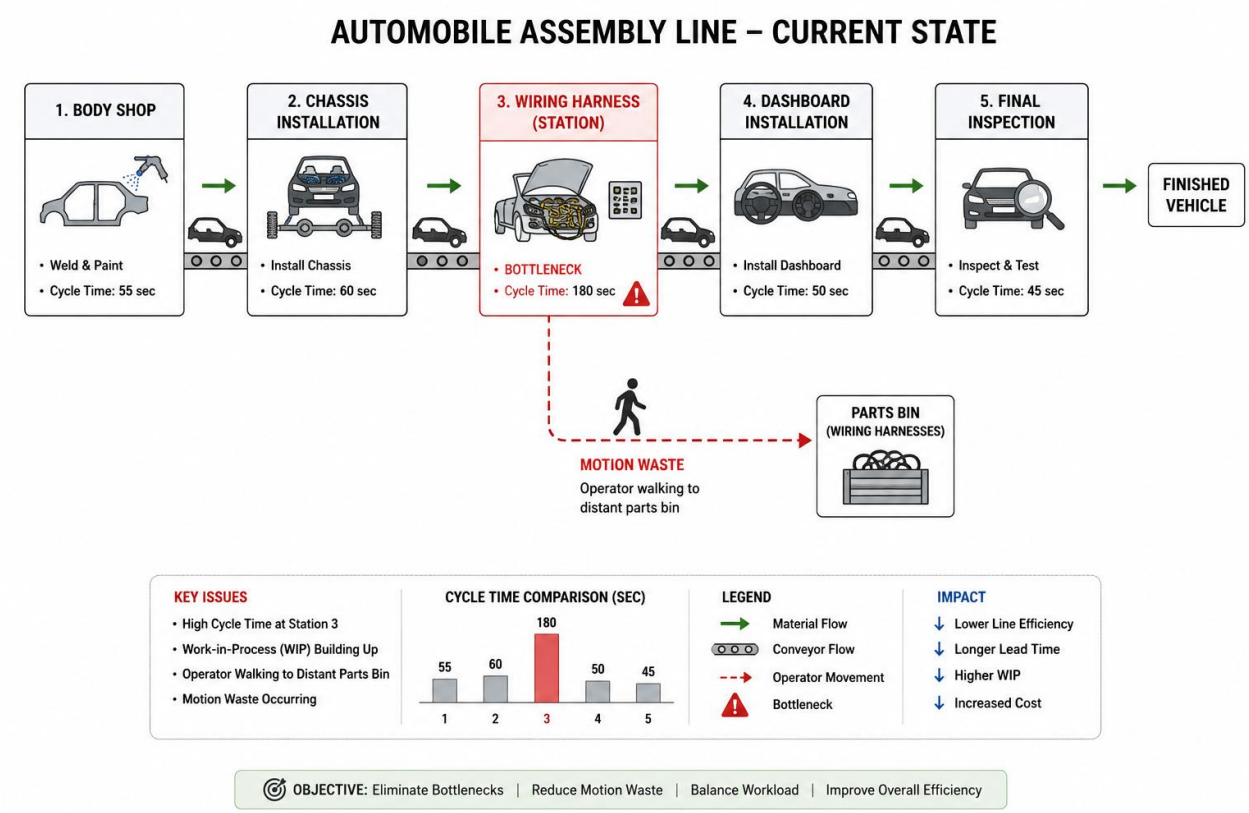
The three-week observation phase generated a massive dataset. However, a filtering of this data immediately highlighted distinct operational patterns that contradicted management's initial assumptions.

8.1 The Reality of the Line Pace

The most critical observation was that the line was *not* universally slow. The time data revealed a severely uneven line balance. A small number of overloaded stations were forcing the entire line to run below its true potential, while several faster stations were leaving workers with idle time that they spent simply waiting for the next vehicle to arrive.

8.2 Bottleneck Station 1: Wiring Harness Fitting

Operators at the Wiring Harness station consistently failed to beat the conveyor pace. Observational data revealed major motion waste. The operators were forced to walk approximately two meters away from the vehicle to a supply bin *twice* per car. Furthermore, they were repeatedly bending down to floor-level shelves to retrieve small plastic connectors. Additionally, the complex wiring harnesses were delivered in tangled bundles, requiring the operator to spend precious seconds manually untangling wires before any actual installation work could begin.



8.3 Bottleneck Station 2: Dashboard Fitting

The Dashboard Fitting station presented severe ergonomic and procedural flaws. The dashboard panel is a large, cumbersome, and moderately heavy component. The operators were required to lift and hold the dashboard in perfect alignment with one hand, while simultaneously attempting to insert and tighten fixing screws with the other hand. Because no mechanical support or jig was provided, operators frequently struggled, dropped screws, or had to shout for temporary assistance from a neighboring station, causing cascading delays.

9. In-Depth Analysis

With the raw data collected, the engineering team applied the analytical tools of Method Study and Time Study to quantify the inefficiencies.

9.1 Method Study Analysis (Identifying 'Muda' or Waste)

The Method Study confirmed that a significant portion of the work being done at the bottleneck stations added absolutely no value to the final automobile.

- **Wiring Harness:** Walking two meters and bending to the floor are classified as "Motion Waste." Untangling wires is classified as "Over-processing Waste." These actions are

purely the result of a workstation layout that was designed eight years ago and never updated or optimized.

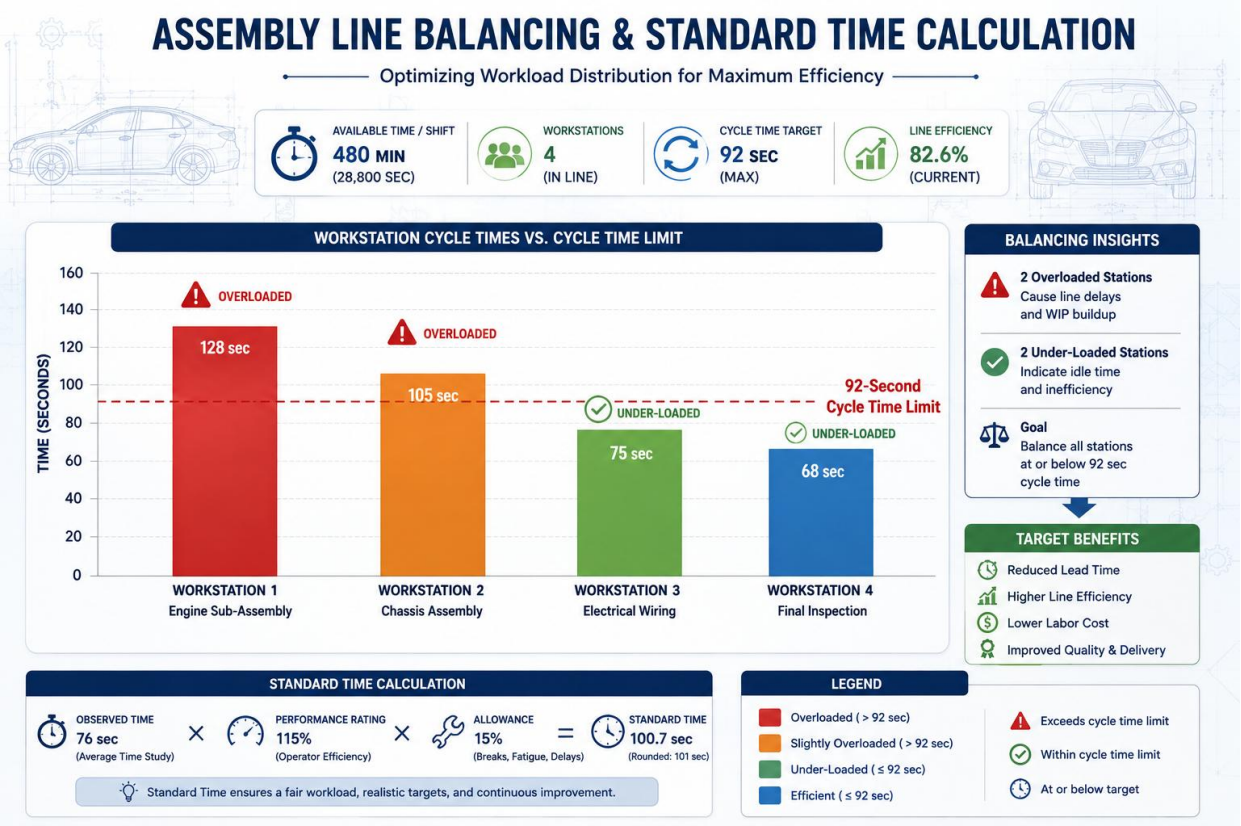
- **Dashboard Fitting:** The lack of a supporting jig is an engineering failure. The awkward one-handed grip not only slows the process down drastically but induces severe muscular fatigue in the operator's shoulder and back, directly explaining the high absenteeism rate reported by HR for this specific zone.

9.2 Time Study Calculations

The team translated the stopwatch readings into standardized metrics. To find the Standard Time, they first calculated the Normal Time (Observed Time multiplied by the operator's Performance Rating) and then added a standard industry allowance of 15% to account for rest, personal needs, and unavoidable factory delays.

The critical comparative data against the 92-second conveyor Cycle Time is presented below:

Station Name	**Observed Time (sec)**	**Performance Rating (%)**	**Normal Time (sec)**	**Standard Time (sec, +15% Allowance)**	**Variance to 92s Cycle Time**
Wiring Harness Fitting	96	105%	101	**116**	**- 24 seconds (Overloaded)**
Dashboard Fitting	101	100%	101	**116**	**- 24 seconds (Overloaded)**
Seat Fitting (Comparison)	78	100%	78	**90**	+ 2 seconds (Balanced)
Wheel Fitting (Comparison)	70	100%	70	**81**	+ 11 seconds (Under-loaded)



9.3 Line Balancing Analysis

The standard time table explicitly reveals the mathematical root of the plant's failure. The conveyor dictates that a car must be finished every 92 seconds. However, the Wiring Harness and Dashboard stations fundamentally require 116 seconds to complete their work using the current methods.

This results in a 24-second deficit on *every single car*. Over an eight-hour shift (28,800 seconds), a 24-second delay per unit drastically reduces the total number of units that can pass through the bottleneck. The analysis also confirmed that at least three neighboring stations (such as Wheel Fitting at 81 seconds) possessed 15% to 20% spare capacity that was completely unutilized.

9.4 The Compounding Impact of the Problem

Because the line is rigidly conveyor-paced, the 24-second shortfall at the two bottleneck stations creates a domino effect. When a car passes the Dashboard station incomplete, it travels down the line missing critical fasteners. Downstream stations cannot attach secondary trim panels because the primary dashboard is loose. This results in the vehicle reaching the Final Line

requiring heavy rework, forcing management to authorize expensive overtime hours and lowering the overall quality standard of the automobile.

10. Summary of Findings

Based on the rigorous application of Work Study principles, the following factual findings are formalized:

1. **Uneven Line Balance:** The assembly line is not generally slow; it is severely unbalanced. The entire production rate is being choked by two specific bottlenecks: Wiring Harness Fitting and Dashboard Fitting.
2. **Mathematical Shortfall:** Both bottleneck stations require 116 seconds of Standard Time, heavily exceeding the fixed 92-second conveyor cycle time.
3. **Root Cause of Delay:** The 24-second overload at these stations is strictly caused by poor spatial layout, lack of proper tooling (jigs), and avoidable physical motions, *not* by a lack of worker skill or effort.
4. **Under-Utilized Assets:** Several neighboring stations (e.g., Wheel Fitting) possess significant spare capacity (up to 11 seconds of idle time per cycle) that is not being productively leveraged.
5. **Ergonomic Failure:** The high absenteeism and fatigue complaints correlate perfectly with the two overloaded stations, proving a direct link between poor Method Design and human physical strain.
6. **Lack of Continuous Improvement:** The facility has suffered from a lack of routine operational auditing; no structured Method or Time Study had been executed since the plant opened eight years ago.

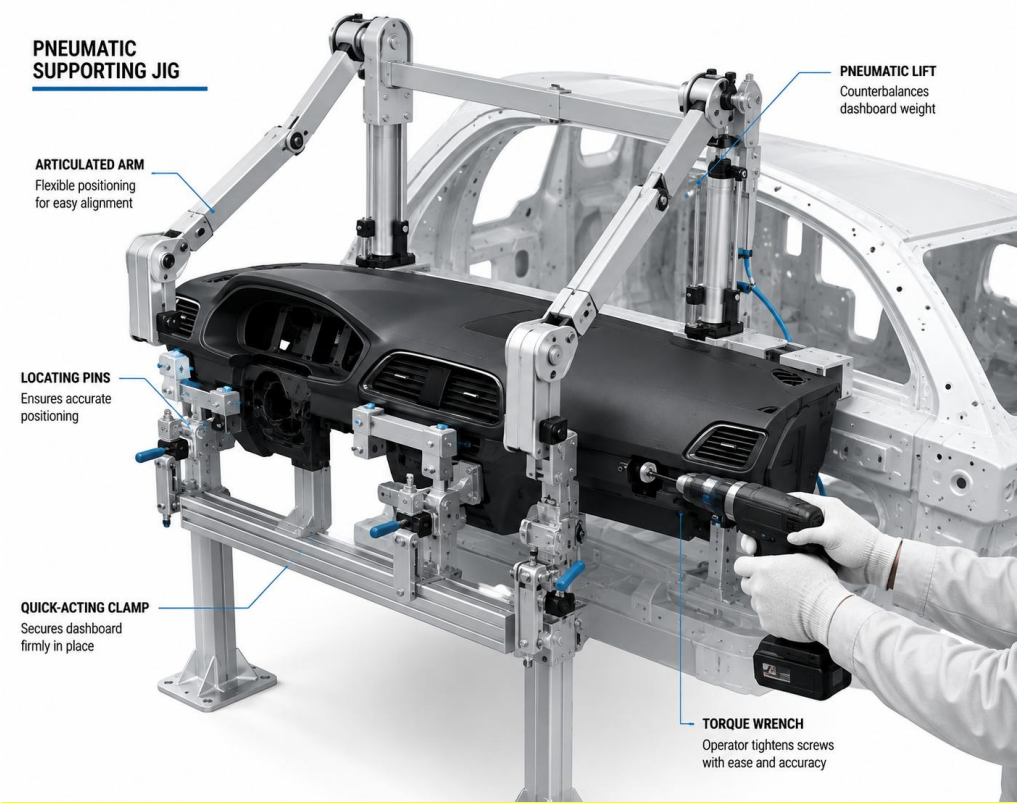
11. Strategic Recommendations and Implementation Plan

To resolve the 24-second deficit and achieve the target of 260 vehicles per shift, Trident Motors must implement the following operations management interventions. These recommendations focus on working smarter, requiring minimal capital investment.

11.1 Immediate Ergonomic and Layout Redesign

- **Wiring Harness Station:** Immediately redesign the spatial layout. The parts bin must be moved directly adjacent to the assembly point (eliminating the two-meter walk). Small connectors must be relocated from floor-level shelves to angled, waist-height gravity-feed racks. This will instantly eliminate the bending and walking motion waste, shaving an estimated 12 seconds off the observed time.

- **Dashboard Fitting Station:** The engineering department must immediately fabricate and install a pneumatic or mechanical supporting jig. This jig will hold the heavy dashboard panel perfectly in place against the chassis. This allows the operator to use both hands simultaneously with electric torque wrenches to secure the fasteners, eliminating the awkward one-handed struggle and reducing physical strain. This is estimated to save 15 seconds per cycle.



11.2 Pre-Kitting and Upstream Preparation

- **Harness Preparation:** Institute a "Pre-Kitting" protocol. Instead of delivering tangled bundles of wires to the line, a sub-assembly worker off-line should untangle, sequence, and place the harnesses on a shadow board. When the harness reaches the main line operator, it is ready for instantaneous installation, completely removing the "untangling" time from the critical path of the conveyor cycle.

11.3 Assembly Line Rebalancing

- **Work Element Shifting:** Even with layout improvements, the workload must be equalized. Management must conduct a Line Balancing exercise to identify small, easily

transferable micro-tasks (e.g., snapping on a plastic trim cover, or routing a single secondary wire) from the Wiring Harness and Dashboard stations. These 5-to-10-second tasks must be permanently reassigned to the under-loaded neighboring stations (such as Wheel Fitting), bringing all stations as close to the 92-second mean as possible.

11.4 Long-Term Operational Hygiene

- **Routine Work Study Audits:** Management must abandon the "set it and forget it" mentality. A policy must be established requiring a full Time and Method Study review of the entire line annually, or immediately following the introduction of any new vehicle trim variant.
- **Cross-Training and Rotation:** To combat repetitive strain injuries and boredom, operators should be cross-trained and rotated between fast-paced and moderately-paced stations every four hours.

12. Conclusion

The operational analysis of the Trident Motors Assembly Division serves as a powerful testament to the enduring relevance of traditional Work Study techniques. Prior to this investigation, management was attempting to solve a production deficit through guesswork, assuming that adjusting conveyor speeds or pressuring workers would magically yield the missing 55 vehicles per shift.

By applying the precise, empirical frameworks of Method Study and Time Study, the engineering team transformed a chaotic factory floor into a solvable mathematical equation. The study proved that an assembly line can appear incredibly busy while simultaneously failing to meet its objectives due to an uneven distribution of work content. The identification of the Wiring Harness and Dashboard stations as severe bottlenecks—consuming 116 seconds against a 92-second limit—isolated the exact source of the factory's failure.

The recommended interventions highlight the elegance of Operations Management: by simply moving a parts bin, providing a mechanical holding jig, untangling wires in advance, and shifting minor tasks to idle workers, the plant can entirely eliminate the 24-second deficit. These low-cost, high-impact solutions directly address the root causes of the delay while drastically reducing the physical toll on the human workforce. Ultimately, this case study demonstrates that continuous, data-driven observation of how work is actually performed is the only reliable pathway to unlocking true manufacturing productivity.

13. References

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